

1. Annotated Bibliography

Arntz, A., van den Hoorn, M., Cornelis, J., Verheul, R., van den Bosch, W. M., & de Bie, A. J. (2003). Reliability and validity of the borderline personality disorder severity index. *Journal of personality disorders*, 17(1), 45–59.
<https://doi.org/10.1521/pedi.17.1.45.24053>

This study was conducted on borderline personality disorder patients and those suffering from various other personality disorders, both assessed via the SCID-5-PD previously. Researchers employed the Borderline Personality Disorder Severity Index (BPDSI). Comparisons of the two groups validates the BPDSI as a reliable assessment tool for measuring BPD frequency and severity as the BPDSI was able to distinguish between those who had BPD and other disorders. Knowing this, we can use this method during our procedure.

Bohus, M., Kleindienst, N., Limberger, M. F., Stieglitz, R. D., Domsalla, M., Chapman, A. L., Steil, R., Philipsen, A., & Wolf, M. (2009). The short version of the Borderline Symptom List (BSL-23): development and initial data on psychometric properties. *Psychopathology*, 42(1), 32–39. <https://doi.org/10.1159/000173701>

This study validates the Borderline Symptom List (BSL-23) as an efficient self-rating measure for assessing BPD symptoms. Researchers conducted a study on a total of 659 BPD participants and made comparisons between previously established assessment tools (the BSL-95) and the concise BSL-23. Knowing this, we can use this method during our procedure for increased efficiency while still maintaining sufficient assessment.

Goshen, I. (2014). The optogenetic revolution in memory research. *Trends in Neurosciences*, 37(9), 511–522. <https://doi.org/10.1016/j.tins.2014.06.002>

This paper discusses the functionality of optogenetics in terms of its use in memory and memory manipulation. Since the technique requires entering the brain, it is considered invasive. Raising the invasive nature of the treatment, we can use this knowledge in our project as a part of the ethical concerns and discussion.

Kohara, K., & Okada, M. (2023). Single-Cell Labeling Strategies to Dissect Neuronal Structures and Local Functions. *Biology*, 12(2), 321.
<https://doi.org/10.3390/biology12020321>

This review discusses how complexities in neuronal structures and functions in the brain make isolating a single memory challenging. Kohara and Okada examine techniques for single-cell labelling, including optogenetics.

Liu, X., Ramirez, S., Pang, P. T., Puryear, C. B., Govindarajan, A., Deisseroth, K., & Tonegawa, S. (2012). Optogenetic stimulation of a hippocampal engram activates fear memory recall. *Nature*, 484(7394), 381–385.
<https://doi.org/10.1038/nature11028>

This study demonstrates how memories are encoded by neurons, and that by using optogenetics to reactivate a specific group of neurons in mice, fear memories can be recalled—and later recontextualized—in different settings.

Soloff P. H. (2021). Bridging the Gap Between Remission and Recovery in BPD: Qualitative Versus Quantitative Perspectives. *Journal of personality disorders*, 35(1), 21–40. https://doi.org/10.1521/pedi_2019_33_419

This article examines the effectiveness of current treatments for BPD through observing 150 BPD individuals for an average of 10 years. The results highlight the gap between symptom remission and full psychosocial recovery.

Sosic-Vasic, Z., Eberhardt, J., Bosch, J. E., Dommès, L., Labek, K., Buchheim, A., & Viviani, R. (2019). Mirror neuron activations in encoding of psychic pain in borderline personality disorder. *NeuroImage Clinical*, 22, 101737-101737.
<https://doi.org/10.1016/j.nicl.2019.101737>

This article highlights the neuronal regions associated with the emotional instability and psychic pain experienced by patients with BPD. Studies such as this can be used to decide what neurons to target when using a treatment such as optogenetics.

Yen S, Peters JR, Nishar S, et al. Association of Borderline Personality Disorder Criteria With Suicide Attempts: Findings From the Collaborative Longitudinal Study of Personality Disorders Over 10 Years of Follow-up. *JAMA Psychiatry*. 2021;78(2):187–194. doi:10.1001/jamapsychiatry.2020.3598

This study demonstrates the significance of BPD in association with suicide attempts and mortality. This factor should be considered when understanding the importance of preventative care as well as efficacy, especially when concerning a treatment that could cause irreversible changes, such as optogenetics. Additionally, the risk of suicide could be used as a criteria for choosing participants for the study.